Printed Electronics Based Optical Unclonable Label for Artificial Intelligence Authentication

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***Abstract*—Printed electronics (PE) products have been widely used in diverse applications, ranging from consumer electronics to automotive and healthcare. The mass manufacturability, flexibility, and low cost of lightweight PE products increase their vulnerability to counterfeiting, allowing fake or substandard goods to infiltrate the supply chain. This poses significant risks to both consumers and businesses. In this paper, we propose to print an ultra-low-cost optical unclonable label (OUL) for the authentication of PE or other inexpensive products. The label pattern is easily designed and printed in a negligibly small area. Random fusions among multiple adjacent ink droplets are explored to form unique and unpredictable patterns. The overall fabrication is compatible with the typical PE process. The inherent randomness of production variations in OULs allows for an extremely high encoding capacity of up to 1.59×10^{3413} with a footprint of only 4×4 mm^2. We introduce an artificial intelligence (AI) system to authenticate the printed OULs under diverse imaging conditions. The authentication among six categories and 6,000 labels can achieve an ideal accuracy. The authentication accuracy of 300 categories can still achieve an average recognition accuracy of 0.97. Beyond exhibiting robust reliability against bending, thermal fluctuations, humidity, and one-year accelerated aging, the printed OULs are highly resilient to both physical replication and machine-learning-based forgery attempts. To enable practical deployment, the lightweight authentication model is implemented as a smartphone application, allowing users to effortlessly capture OUL images and obtain reliable on-site verification.**

***Index Terms*—anti-counterfeiting, AI authentication, optical unclonable label (OUL), printed electronics (PE).**

I. Introduction

As an additive manufacturing technology, printed electronics (PE) enables electronic devices and systems to be produced using conventional printing techniques [1]. PE products can hence be fabricated on a large scale at low cost while offering customizability, mechanical flexibility, and environmental friendliness [2]. These merits promote PE to be adopted in various fields, including the Internet of Things (IoT) [3], smart sensors [4], smart packaging [5], smart labels [6], and ingestible health diagnostic equipment [7]. Notably, PE products excel in emerging applications like Internet of Disposable Things (IoDT) [8] and Electronic Skin (E-Skin) [9]. The global PE market size is projected to reach $75.05 billion by 2026, representing a compound annual growth rate of 22.4% from 2024 to 2032 [10].

Lightweight PE devices offer significant benefits, yet they are vulnerable to counterfeiting. Such counterfeit devices can disseminate false information, deny service, or be used as weapons to infiltrate other networks. This poses serious threats and potentially catastrophic consequences in their respective applications [11]. To mitigate this risk, several anti-counterfeiting solutions, such as copy-resistant barcodes [12], laser ablation [13], electro-chemical jet [14], and radio frequency identification (RFID) [15], have been developed for authenticating PE products. However, these countermeasures are not inherently lightweight and lack the capability to provide unique authentication under identical tag patterns.

Lightweight authentication is essential for enabling practical and feasible security in a wide range of PE applications. Physical unclonable functions (PUFs) shed new light on anti-counterfeiting solutions by generating unique identifiers for each product based on inherent manufacturing variations [16]-[21]. These PUF-based IDs demonstrate strong uniqueness and unclonability, making them ideal for authentication purposes [22]-[26]. Several electronic PUFs have been developed for PE products, such as printed memory PUF [23], RO-PUF [24], and PE-PUF [25]. However, their circuit complexity often leads to yield issues when integrated into lightweight PE devices.

Recent research has explored physical unclonable labels using optical properties of multiple specific materials, such as quantum dots [27], amorphous photonic structures [28], structure color [29], and random fractal patterns [30], colloidal nanowires [37], and perovskite [38]. These labels provide high information entropy and eliminate the need for custom circuitry, thereby reducing costs and enhancing application flexibility. However, they rely on specialized materials, complex fabrication processes, and professional optical readers [25]-[32], [37]. Moreover, practical limitations, including the high cost of Au networks [30], quantum dot toxicity [33], the environmental sensitivity of structural color [34], and optical scattering [39], further impede their adoption in PE devices.

Efficient authentication of PUF-based IDs remains another key challenge. Conventional fingerprint verification requires feature extraction, database storage, and linear-time matching, which become increasingly burdensome as the number of enrolled fingerprints grows. Artificial intelligence (AI) presents a promising alternative [27]-[30]. It leverages data-driven learning to build efficient decision-making models. AI achieves

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a better balance among efficiency, cost, and security compared to traditional approaches. It has been successfully applied in biometric identification and smart control systems for authenticating unclonable labels [27]-[29], [38], [40]. Recent works have demonstrated that deep learning models, ranging from lightweight PaCoNet [38] to ResNet [27], [40] can facilitate convenient end-user authentication. However, maintaining high accuracy without compromising efficiency remains a major bottleneck.

In this work, we investigate how ink droplets of aqueous silver nanoparticles (AgNPs), a common printing material, form diverse and inherently unpredictable patterns when printed onto a typical polyethylene terephthalate (PET) film substrate. These patterns form stochastically even under uniform printing conditions, yet exhibit sufficient information entropy to serve as natural physical fingerprints. Inspired by this finding, we propose a new optical unclonable label (OUL) fabricated via the standard printing process, designed for AI-driven authentication. We develop a feature extraction method to identify the region of interest (ROI) from each OUL and employ an advanced AI system to learn the ROI patterns and build a robust authentication model. A smartphone application is also implemented to facilitate practical, real-world verification. To validate the proposed method, 300 OUL samples are printed and evaluated under multiple authentication scenarios, with the results confirming the feasibility of our scheme.

The contributions of this work are summarized as follows:

(1) We design and fabricate a novel OUL that is fully compatible with standard printing processes. The proposed label is cost-effective, miniaturized (4×4 mm^2), and flexible, allowing seamless integration into the margins of PE products. Authentication can be performed with a common camera, ensuring wide applicability without specialized equipment.

(2) We establish a foundational framework for optimizing and quantitatively evaluating the OUL's uniqueness and security. A derived formulation guides the setting of printing parameters to maximize randomness. Metrics such as image entropy, Quantity of Information under varying Printing parameters (*QIP*) are introduced, demonstrating an exceptionally high encoding capacity 1.59×10^{3413}, which underscores strong anti-counterfeiting capabilities.

(3) We implement a practical, AI-powered authentication system, complemented by a functional smartphone application. This end-to-end solution incorporates an accurate and rapid AI model. Its implementation in a user-friendly app validates the scheme's robustness, convenience, and real-world viability.

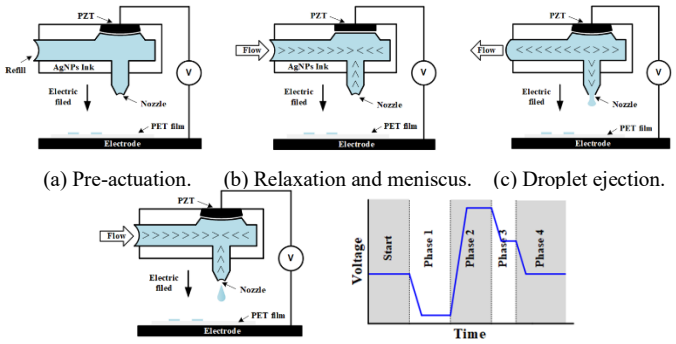
The rest of this paper is organized as follows. Section II reviews existing lightweight PE authentication techniques. Section III introduces the preliminaries of inkjet printing. The design of the OUL and AI authentication system is detailed in Section IV. Experimental results are analyzed in Section V. The paper is concluded in Section VI.

II. Related Work

OULs are well-suited for PE applications due to their lightweight nature. Matsumoto et al. [32] developed specialized marking processes to generate unique nanoscale random patterns as device-specific fingerprints. However, their dependence on expensive dedicated etching equipment limits their applicability in PE systems. Benedikt et al. [26] proposed to use the inherent surface texture of injection-molded plastic components as an unclonable feature to reduce authentication costs. Sharma et al. [35] extended this concept by adopting image texture instead of microstructure and established a low-cost authentication framework. Similarly, Takahashi et al. [36] introduced the "FIBAR" method to capture naturally formed uniqueness on metal parts. These methods avoid extra serialization costs and confirm the feasibility of optical PUF-based item authentication. Notably, Erozan et al. proposed an image-based PUF (iPUF) [31] for PE devices, which generates fingerprints by capturing optical variations in printed patterns. However, since functionally similar PE devices often share structural similarities, the limited shape variations constrain information entropy and lead to weak inter-device discrimination. To address this, researchers have explored OULs using specialized materials such as quantum dots [27], amorphous photonic structures [28], structural color [29], random fractal structures [30], colloidal nanowires [37], and perovskite [38]. For instance, Liu et al. [27] developed a quantum dot fluorescent anti-counterfeiting label that leverages stochastic pinning points of ink droplets to form unclonable flower-like patterns. He et al. [28] designed a structural-color label based on amorphous photonic structures, enabling three-level security. Li et al. [29] introduced another structural-color label using cross-linked short-range ordered arrays of polymeric nanospheres, achieving full-color PUF. Cao et al. [37] showed that spin-coating colloidal silver nanowires form random network structures suitable for PUF-based triple-key authentication. Han et al. [38] fabricated perovskite fluorescent anti-counterfeiting labels via surface-tension-confined droplet arrays on lyophilic-lyophobic patterned substrates. Kim et al. [39] exploited random surface scattering to realize multi-level optical PUFs that enable dynamic hierarchical key generation from a single tag. More recently, Wei et al. [40] extended optical PUF technology to wearable platforms.

Traditional authentication of optical labels depends on pattern recognition and comparative analysis, which are computationally intensive and often lack satisfactory accuracy. AI has introduced new possibilities in image processing and pattern recognition, offering improvements in both accuracy and efficiency. However, existing AI-based approaches still have limitations: Liu et al. [27] and He et al. [28] reported inconsistent authentication accuracy across different label categories. This indicates poor generalization of label-dependent models. Sun et al. [30] proposed a dynamic AI authentication method that achieved zero false positives. However, it relied on feature similarity analysis and required storing large-scale original datasets. Moreover, existing AI authentication schemes [27]-[30], [38], [40] have been validated at modest scales (typically fewer than ten categories),

and employ only limited data augmentation during training, which may yield a misleading perception of model robustness. Real-world scenarios are significantly more complex and necessitate extensive data augmentation to simulate realistic variations. Given that PE products are typically mass-produced, scaling up label deployment presents considerable challenges for AI model training. To overcome these issues, this work utilizes a standard PE fabrication process to print a new OUL for lightweight AI authentication.



(a) Pre-actuation. (b) Relaxation and meniscus. (c) Droplet ejection.

(d) Stabilization. (e) Waveform.

Fig. 1. The jetting process and standard waveform of inkjet printing.

III. Preliminaries

Inkjet printing is a digitally controlled, non-contact, drop-on-demand technique that enables the direct deposition of nano-sized liquid solutions onto flexible or rigid substrates [41]. The printing device typically consists of a capillary (the pumping chamber) headed by a nozzle with several tens of micrometers in diameter, and a piezoelectric actuator. As illustrated in Fig. 1, the jetting process, driven by a standard waveform optimized for the Dimatix Model Fluid, includes four phases. (1) Pre-actuation. As shown in Fig. 1(a), the lead zirconate titanate (PZT) bimorph undergoes a slight deflection before the main drive pulse begins. (2) Relaxation and meniscus formation. As shown in Fig. 1(b), when the voltage returns to zero, the PZT actuator relaxes, drawing ink into the chamber and forming a meniscus at the nozzle. (3) Droplet ejection. As shown in Fig. 1(c), an applied voltage pulse causes the PZT to deflect, compressing the pumping chamber and generating pressure to eject a droplet. (4) Stabilization. As shown in Fig. 1(d), the PZT voltage resets to its bias level, and its motion is controlled to ensure clean droplet breakoff. Fig. 1(e) presents a typical driving waveform, structured around these four phases and optimized for the Dimatix Model Fluid. Upon impacting the PET substrate, the aqueous ink spreads under the influence of surface tension, forming a circular dot of diameter *DR*. To fabricate the desired pattern, the Drop Spacing (*DS*), defined as the center-to-center distance between adjacent droplets, can be adjusted to control the deposition positions of the droplets.

IV. Proposed OUL-based Authentication Scheme

This section firstly elaborates on the fabrication process of the proposed OULs. A comprehensive feature extraction methodology is then presented, which is essential for authentication performance. Subsequently, an AI-based authentication scheme is introduced. Finally, a set of evaluation metrics is defined to assess the performance of both the OULs and the AI authentication system.

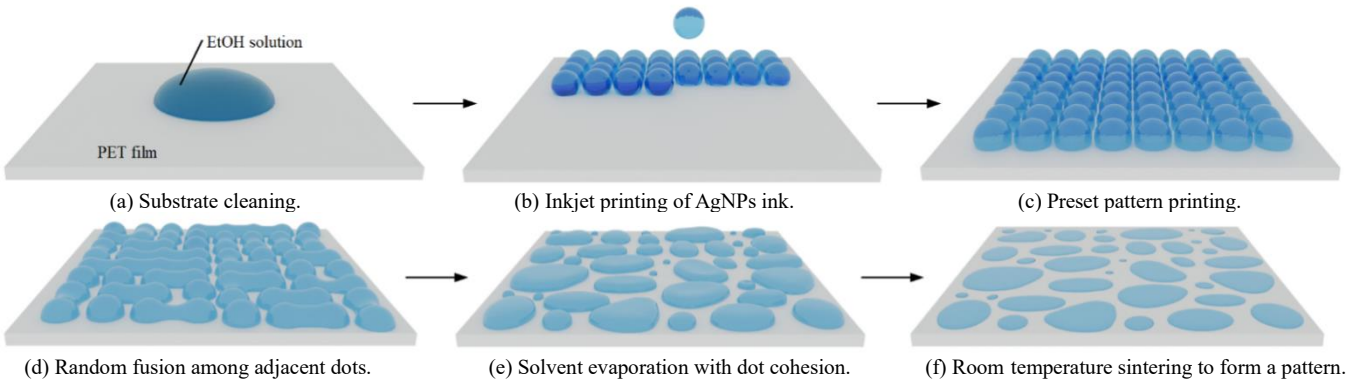
A. Fabrication of OULs

1) Material Selection

In this work, AgNPs ink is selected as the printing material due to its low-temperature sintering capability, excellent flex resistance, high stability, and tunable viscosity [42]. The AgNPs ink used is synthesized via a well-established chemical route [43], [44]. It is particle-free, with a solid content of 8wt.%. The solvent is a blend of ethylene glycol and methanol: ethylene glycol promotes the dispersion of AgNPs, while methanol helps adjust viscosity and surface tension. A transparent, glossy PET film is used as the substrate due to its excellent mechanical and optical properties. A Dimatix DMP-2850 inkjet printer is employed as the material precipitation instrument.

2) OULs Fabrication Process

Fig. 2 illustrates the formation processes of the printed OUL by inkjet printing. To ensure strong adhesion, the PET film surface is first cleaned with a small amount of ethanol (EtOH) solution and dried, as shown in Fig. 2(a). The AgNPs ink is then deposited onto the PET film through inkjet printing, as depicted in Fig. 2(b) and Fig. 2(c). After printing, adjacent droplets coalesce randomly, forming irregular patterns, as shown in Fig. 2(d). As the solvent evaporates, the dots further merge and solidify, as shown in Fig. 2(e). Over time, multiple isolated fused droplets continue to dry and stabilize, ultimately forming the final printed OUL pattern after two days at room temperature, as shown in Fig. 2(f).



(a) Substrate cleaning. (b) Inkjet printing of AgNPs ink. (c) Preset pattern printing.

(d) Random fusion among adjacent dots. (e) Solvent evaporation with dot cohesion. (f) Room temperature sintering to form a pattern.

Fig. 2. Printing processes of the OULs.

This fabrication process is inherently random. Minor variations in factors, such as ink properties, printer operation, cartridge depletion status, ink/substrate batches, environmental conditions, and substrate characteristics, collectively influence the behavior of each droplet. These variations lead to distinct droplet fusion patterns, even when identical voltage waveforms are used. Ultimately, each OUL is formed as a randomized pattern within a 4×4 mm^2 area, providing a unique physical fingerprint that serves as a secure cryptographic ID to identify a device.

3) Formulation of Parameters

The formulation of the drop spacing *DS* parameter is critical, as it governs the random coalescence of adjacent ink droplets, as shown in Fig. 3. When *DS* is much larger than *DR*, the printed dots remain isolated, forming discrete patterns with negligible random variation, as shown in Fig. 3(a). Conversely, if *DS* is much smaller than *DR*, as shown in Fig. 3(b), multiple circular dots extensively merge into solid patterns, which substantially reduces the overall pattern entropy. The optimal randomization is achieved when *DS* is approximately equal to *DR*, as shown in Fig. 3(c), which promotes sufficient droplet interaction. Therefore, *DR* must be carefully characterized to properly configure the *DS* value for generating OULs with high randomness.



(a) *DS* > *DR* (b) *DS* < *DR* (c) *DS* ≈ *DR*

Fig. 3. Random fusion among multiple adjacent ink droplets in printed patterns at different *DS*.

During the printing process, multiple droplets are ejected from the nozzle and deposit onto the PET film, coalescing to form a circle-like drop. Let the thickness and diameter of the resulting dot be denoted as h and DR, respectively. The total volume of the dot, V_d, can be expressed as:

$$V_d = nV_0 = \pi \cdot \left(\frac{DR}{2}\right)^2 \cdot h \tag{1}$$

where V_0 is the volume of a single droplet, and n is the number of droplets that merge into one dot.

According to the Young-Laplace equation [45], at equilibrium, the capillary pressure induced by surface tension is balanced by the liquid's internal pressure. This relationship is given by (2):

$$\Delta P = \gamma \left(\frac{1}{r_1} + \frac{1}{r_2}\right) = \frac{4\gamma \cdot cos\theta}{DR} \tag{2}$$

where ΔP denotes the pressure difference across the liquid-air interface, γ is the surface tension of the ink, r_1 and r_2 are the principal radii of curvature, and θ denotes the contact angle determined by the ink viscosity.

As the internal droplet pressure is determined by its height h, (2) can be rewritten as:

$$\Delta P = \frac{4\gamma \cdot cos\theta}{DR} = \frac{\rho_{ink} g h}{2} \tag{3}$$

where ρ_{ink} denotes the ink density and g denotes the gravitational acceleration.

The height h can then be computed by:

$$h = \frac{8\gamma \cdot cos\theta}{DR \cdot \rho_{ink} g} \tag{4}$$

According to (1) and (4), *DR* can be computed as:

$$DR = \frac{V_d \cdot \rho_{ink} g}{2\pi\gamma \cdot cos\theta} \tag{5}$$

As *DR* depends on multiple variants that exhibit inherent variations, a scaling coefficient k is introduced to account for these fluctuations and yield a more accurate expression for *DR*:

$$DR = k \cdot \frac{V_d \cdot \rho_{ink} g}{2\pi\gamma \cdot cos\theta} \tag{6}$$

In the printing system, *DS* is governed by the equation $DS = 254\sin\theta_h$, where θ_h represents the cartridge mounting angle. Therefore, the value of *DR* can be utilized to determine the appropriate θ_h required to achieve a target *DS*. Based on the material properties, printer configuration, and preliminary experimental results, the assigned values of all relevant parameters are summarized in Table I.

TABLE I. PARAMETER SETTINGS

No.	Features	Symbols	values
1	single droplet volume	V_0	10pl
2	#droples/drop	n	1000
3	ink density	ρ_{ink}	0.852×10^3 kg/m^3
4	gravitational acceleration	g	9.81m/s^2
5	$r_1 = r_2$	$r_1 = r_2$	*DR*/2
6	surface tension	γ	26.7±0.5 dyne/cm
7	contact angle	θ	0
8	scaling coefficient	k	150
9	dot diameter	*DR*	41.38μm
10	drop spacing	*DS*	45μm
11	cartridge mounting angle	θ_h	7.6°

B. ROI Extraction

After the printed label patterns are finalized on the substrate, they are imaged and processed to extract ROIs for dataset construction. The procedure consists of the following five steps:

Step 1. Imaging. As shown in Fig. 4(a), the OULs are captured using a microscope and transmitted via fiber optics to a computer for observation.

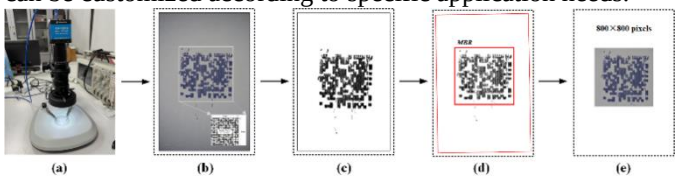
Step 2. Grayscaling and binarization. The original color images, as illustrated in Fig. 4(b), are first converted into grayscale. Binarization is then applied to convert grayscale pixels into values of 0 or 255, substantially reducing data volume and generating a clear black-and-white image, as shown in Fig. 4(c). Given the high contrast between the ink droplets and the background, a threshold value of 119 is used to effectively segment the ink droplets' regions from the background.

Step 3. Pattern detection. The minimum bounding rectangle (MBR) algorithm is applied to the binarized image to identify the contour with the largest droplet area, which is designated as the MBR pattern.

Step 4. Image alignment. The four vertex coordinates of MBR are recorded to define the MBR pattern space. Based on the spatial relationship between the MBR pattern space and the target space in the horizontal direction, a transformation matrix is computed. This matrix defines the mapping rules for rotating

the pattern from the original space to the target space, thereby compensating for rotational effects, as shown in Fig. 4(d).

Step 5. ROI cropping. To meet the input requirements of the AI authentication network, a uniformly sized region of 800×800 pixels is cropped out from the center of the aligned pattern as the ROI. This ROI serves as the authentication OUL sample, as shown in Fig. 4(e). Both the OUL pattern and its dimensions can be customized according to specific application needs.



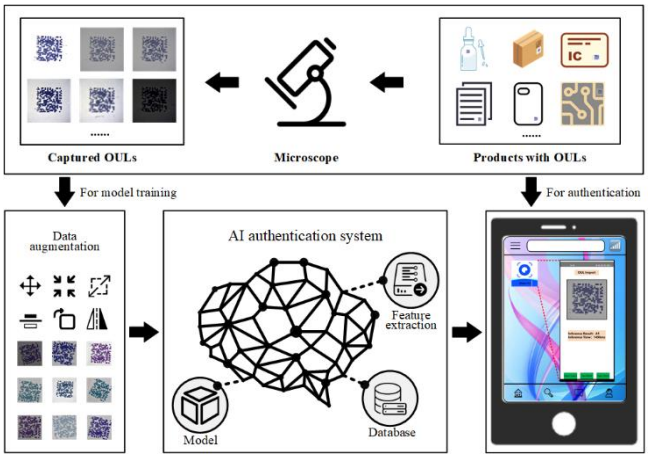
(a) Imaging. (b) Original. (c) Binary. (d) Image alignment. (e) ROI cropping.

Fig. 4. ROI extraction of the printed OUL.

C. AI Authentication System

The adoption of an AI-driven approach is necessitated by two fundamental limitations of conventional centralized database-based approaches: (1) severe performance and lookup bottlenecks when querying massive, scaled response repositories in real time, and (2) systemic privacy risks associated with exposing or compromising raw physical label profiles. Our framework resolves these bottlenecks by embedding instance-level physical feature metrics directly into nonlinear model parameters during training, achieving a 'usable-but-invisible' authentication state. Since the physical fingerprints are bound implicitly within the network weights, the original raw microstructural responses cannot be reverse-engineered even if a local scanning terminal is compromised. Furthermore, fully autonomous offline deployment on consumer edge devices removes remote database dependencies and network latency constraints, significantly enhancing deployment scalability and execution efficiency.

The authentication algorithm extracts grayscale contrast from microscopic images, which arises from reflectance variations caused by the random silver-particle agglomeration during sintering. This physical randomness ensures both label distinctiveness and environmental stability, as supported by the correlation analysis in Section V-A and the reliability analysis in Sections V-C.



(1) Data preparation. (2) Model training. (3) Label validation.

Fig. 5. AI authentication system.

The proposed AI authentication process, as shown in Fig. 5, can be divided into three phases: (1) Data preparation: The original captured OUL labels are augmented to generate sufficient training data. (2) Model training: The prepared data is used to train the model. (3) Label validation: The AI model verifies the authenticity of product labels.

1) Data Preparation

Sufficient label data are required to train the AI model effectively. We employ multiple data augmentation techniques, including rotation, flipping, shifting, distortion, saturation adjustment, Gaussian noise addition, sharpening, blurring, contrast modification, scaling, and combinations thereof, on the original OULs to create derivative images, as shown in Fig. 6. Both the original OUL data and synthetically generated label data are used in training to enhance the model's generalization capability. Each original OUL and its derivatives form a distinct category. Thus, N original OULs correspond to N classification categories.

2) Model Training

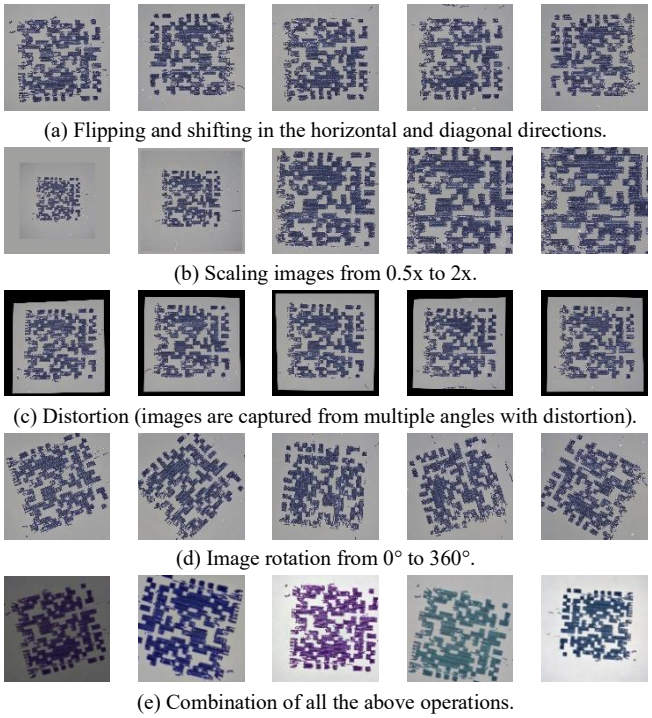
We build a suitable AI network architecture to train a customized classifier capable of learning features from the label dataset. As the winner of the 2016 ILSVRC competition, ResNet-50 [46] has been widely adopted in classification tasks. We therefore employ it as the backbone of our AI system.

To fit the ResNet-50 network input requirements, all images are resized to 512×512 pixels and normalized using statistics from our dataset. A pre-trained model (ImageNet pre-trained weights) is used to initialize the ResNet-50 network, with the cross-entropy loss function, Adam optimizer, and ReLU activation function jointly used during training. The batch size is set to 32. A ReduceLROnPlateau scheduler dynamically adjusts the learning rate, starting from an initial value of 0.01 and reducing by a factor of 0.2 when the validation accuracy fails to improve for 10 consecutive epochs. Training terminates once the accuracy stabilizes over 10 epochs. Complete architectural details, hyperparameter selections, dataset partition mechanics, and rejection criteria are consolidated in S1 of the supplementary material.

To evaluate the model's true generalization capacity, we build and test the network using a strict image-level separation protocol. We follow the process below to build the model. For each of the G genuine labels, we collect $f = 10$ independent photographs under varying physical conditions, denoted as P_1, P_2,…, P_f. The final capture, P_f is reserved exclusively as a completely held-out test set that never participates in any stage of model training or validation. We then performed a (f–1)-fold cross-validation over the remaining (f–1) = 9 captures. For each fold $i \in \{1,\ldots, (f–1)\}$, we train a sub-model M_i using the other (f–2) physical captures and their α augmented derivatives as training data. The capture P_i is used strictly as a validation set to monitor convergence and enforce early stopping. The final evaluation model was subsequently retrained from scratch on the combined dataset of P_1 through $P_{f–1}$ (and their augmented derivatives), with the training epoch and confidence threshold observed across the cross-validation folds, before being

evaluated on the completely unseen test capture P_f. Critically, data augmentation was applied only to the designated training captures within each fold. The validation capture P_i, the held-out test capture P_f, and any of their derivatives were strictly excluded from the training process. This design guarantees zero source-image leakage among the training, validation, and testing phases.

Model performance is evaluated using accuracy and loss metrics to determine when a model has achieved optimal performance or has converged. Although more epochs can improve accuracy, they also increase training time and the risk of overfitting, which degrades generalization. Accordingly, we set the maximum number of epochs to 50 to balance performance gains with the risks of longer training time and poor generalization.



(a) Flipping and shifting in the horizontal and diagonal directions.

(b) Scaling images from 0.5x to 2x.

(c) Distortion (images are captured from multiple angles with distortion).

(d) Image rotation from 0° to 360°.

(e) Combination of all the above operations.

Fig. 6. 30 augmented image examples using different methods.

3) Label Validation

During training, once the evaluation metrics converge or reach a satisfactory level, it indicates that the AI model has successfully learned the features of the OULs. The developer can then save the model parameters and develop an AI authentication application on a smartphone. When an end-user wants to authenticate a product, they can use the app to capture an image of the label and receive a prediction regarding its authenticity. It is important to note that the AI model within the app functions as a black box, making its internal design inaccessible to potential attackers seeking to compromise the authentication system.

D. Evaluation Metrics

In this section, we introduce five metrics to evaluate the quality of the fabricated labels and one metric to assess the performance of the authentication system.

1) Image Entropy

Image entropy measures the average information content in an image. A higher entropy value indicates greater uniqueness among images. We adopt a two-dimensional image entropy [47], denoted as $H_{2D}(X)$, which incorporates both pixel grayscale and spatial information. This entropy is computed using a sliding window centered on each pixel and its neighbors, as defined in (7):

$$H_{2D}(X) = -\sum_{i=0}^{255}\sum_{j=0}^{255} P_{i,j} log P_{i,j} \tag{7}$$

Here, i represents the grayscale value between 0 and 255 of the central pixel, and $j \in [0 \ldots 255]$ denotes the average grayscale of its neighboring pixels within a $k \times k$ window, calculated as:

$$j = \frac{\sum \overline{D}(i,k) - i}{k^2 - 1} \tag{8}$$

where $\sum \overline{D}(i,k)$ is the total grayscale sum of all pixels in the window. The joint probability of $P_{i,j}$ is given by:

$$P_{i,j} = \frac{f(i,j)}{M \cdot N} \tag{9}$$

where M and N denote the image width and height, respectively, and $f(i, j)$ is the frequency of the pair (i, j) in the image. The self-information of (i, j) is defined as $-logP_{i,j}$, with $-logP_{i,j} = 0$ when $P_{i,j} = 0$.

2) *QIP*

To uncouple the influence of printing parameters on information content, we define the quantity of information (*QIP*) under printing parameters as a measure of pattern quality. It is expressed as (10) and its unit is bits/μm.

$$QIP(X) = \frac{H_{2D}(X) * M \times N}{DS} \tag{10}$$

3) *FSIM*

We employ the feature similarity index (*FSIM*) [48] to evaluate the perceptual similarity between two images. *FSIM* combines Phase Congruency (*PC*) and Gradient Magnitude (*GM*), aligning with the Human Visual System (HVS).

The *PC* for each pixel is defined as:

$$PC(\boldsymbol{x}) = \frac{\sum_j E_{\theta_j}(\boldsymbol{x})}{\varepsilon + \sum_n \sum_j A_{n,\theta_j}(\boldsymbol{x})} \tag{11}$$

where $E_{\theta_j}(\boldsymbol{x})$ is the local energy at orientation θ_j, $A_{n,\theta_j}(\boldsymbol{x})$ is the filter response amplitude, and ε is a small constant.

GM is computed as:

$$GM = \sqrt{GM_x^2 + GM_y^2} \tag{12}$$

where GM_x and GM_y represent the horizontal and vertical gradient, respectively.

The similarity measures for *PC* and *GM* are:

$$S_{PC}(\boldsymbol{x}) = \frac{2PC_1(\boldsymbol{x}) \cdot PC_2(\boldsymbol{x}) + T_1}{PC_1^2(\boldsymbol{x}) + PC_2^2(\boldsymbol{x}) + T_1} \tag{13}$$

$$S_{GM}(\boldsymbol{x}) = \frac{2GM_1(\boldsymbol{x}) \cdot GM_2(\boldsymbol{x}) + T_2}{GM_1^2(\boldsymbol{x}) + GM_2^2(\boldsymbol{x}) + T_2} \tag{14}$$

where T_1 and T_2 are stabilization constants.

The overall similarity is:

$$S_L(\boldsymbol{x}) = [S_{PC}(\boldsymbol{x})]^{\alpha} \cdot [S_{GM}(\boldsymbol{x})]^{\beta} \tag{15}$$

The parameters, α and β, adjust the relative importance of *PC* and *GM*. In our experiments, we set $\alpha = \beta = 1$, so $S_L(x) = S_{PC}(x) \cdot S_{GM}(x)$. The maximum *PC* map is defined as:

$$PC_m(\boldsymbol{x}) = max(PC_1(\boldsymbol{x}), PC_2(\boldsymbol{x})) \quad (16)$$

Finally, the *FSIM* score is computed as:

$$FSIM = \frac{\sum_{x\in\Omega} S_L(\boldsymbol{x}) \cdot PC_m(\boldsymbol{x})}{\sum_{x\in\Omega} PC_m(\boldsymbol{x})} \quad (17)$$

4) Hamming Distance

The inter-class fractional Hamming Distance (*inter-HD*) is a standard metric used to evaluate the randomness and uniqueness of PUFs. We adopt *inter-HD* to assess the statistical independence of the proposed printed labels. For *n* independent labels, each generating an *L*-bit binary cryptographic key response B_i, the average *inter-HD* over all pairwise unique combinations is defined as:

$$inter - HD_{avg} = \frac{2}{n(n-1)L} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \sum_{k=1}^{L} B_{i,k} \oplus B_{j,k} \quad (18)$$

where $B_{i,k}$ denotes the *k*-th bit of the *i*-th label's response vector, and $\oplus$ represents the bitwise exclusive-OR (XOR) operation. An ideal, truly random PUF response profile should yield an average *inter-HD* of exactly 0.5, indicating that bit sequences from distinct physical tags are completely uncorrelated and highly distinguishable. Conversely, an average *inter-HD* that deviates significantly from 0.5 implies strong deterministic correlations between labels, indicating poor structural randomness and an increased vulnerability to modeling or machine learning-based prediction attacks.

5) Encoding Capacity

Encoding capacity refers to the number of distinct codes that can be generated from the printed labels. Based on the binary-bit model by Carro-Temboury et al. [49], the encoding capacity for an inkjet droplet pattern is given by:

$$\#codes_F = \left[C \left(1 + L \left(\frac{1}{\sqrt{D}} - 1 \right) \right)^2 + 1 \right]^{D\frac{R^2}{L^2}} \quad (19)$$

where *C* is the number of colors, *D* is the pixel filling density (percentage of preset pattern in the image), *L* is the length of the minimum image unit, and *R* denotes the image resolution.

6) Confusion Matrix

For each queried label, the system performs a two-step pipeline: (1) determining whether the label belongs to the enrolled database, and if so, (2) identifying its specific class among the enrolled categories. This open-set decision is jointly governed by a confidence threshold τ_p (the minimum softmax probability required to accept a predicted class) and a distance threshold τ_d (the maximum allowable cosine distance between the query feature and its nearest class centroid). A label is accepted and classified if and only if both criteria are satisfied. Otherwise, it is rejected as unenrolled.

To evaluate this open-set performance over *P* enrolled and *Q* unenrolled labels, we adopt a modified confusion-matrix framework as defined in Table II. From these metrics, we derive classification *accuracy* (for enrolled samples), and True Rejection Rate (*TRR*, for unenrolled samples) via Eq. (20).

TABLE II. DEFINITION OF OPEN-SET AUTHENTICATION METRICS

No.	Basic metric	Notation	Definition
1	True positive	*TP*	Enrolled samples correctly identified as their true registered class.
2	True negative	*TN*	Unenrolled samples correctly identified and rejected by the system.
3	False positive	*FP*	Unenrolled samples wrongly accepted as registered class.
4	False negative	*FN*	Enrolled samples wrongly rejected as unknown or misclassified.

$$accuracy = \frac{TP}{TP + FN}, \qquad TRR = \frac{TN}{TN + FP} \quad (20)$$

V. RESULTS AND DISCUSSIONS

In the experiment, the proposed method was employed to fabricate 300 original independent OULs. These labels were produced from two printing batches, with 150 OULs per batch. For each physical OUL, 10 repeated microscopic images were captured to characterize real acquisition repeatability. Each image was resized to 800×800 pixels, then divided into a training set, a validation set and a test set, and finally used to train an AI model based on the ResNet-50 architecture. The evaluation was conducted in two stages: first, an assessment of the quality of the printed labels, followed by an analysis of the AI authentication performance of enrolled-label under imaging variation.

A. Evaluation of Printed OULs

Table III presents the image entropy and *QIP* under different *DS* and sliding window size *k*. It can be observed that increasing *DS* leads to higher entropy, which verifies that a *DS* value closer to *DR* enhances label randomness. Furthermore, as *k* increases, entropy also rises since the sliding window more accurately approximates the actual shape of the ink droplets, thereby capturing greater pattern randomness.

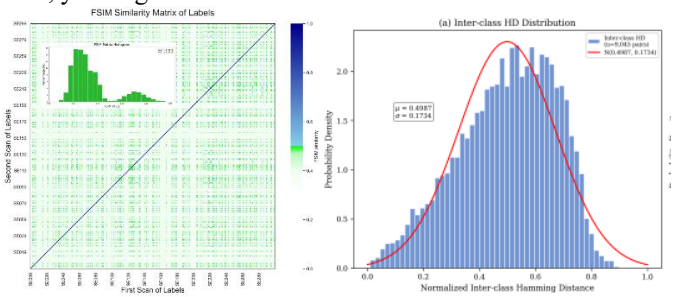
TABLE III. IMAGE ENTROPY AND QIP OBTAINED UNDER DIFFERENT *DS* VALUES

Evaluation	*k*	*DS* = 20μm	*DS* = 35μm	*DS* = 50μm
$H_{2D}(X)$ (bits/pixel)	3	8.806	9.435	12.052
	5	9.595	10.633	13.67
	7	10.157	11.551	14.761
	9	10.597	12.338	15.571
$QIP(X)$ (bits/μm)	3	4.058E+05	2.484E+05	2.222E+05

For the typical case where $k = 3$, $QIP(X)$ decreases with increasing *DS*. This indicates that the entropy does not scale proportionally with *DS*, highlighting the need to balance the entropy and printing parameters to achieve high-quality labels.

Fig. 7(a) presents the correlogram of the *FSIM*-based similarity matrix for all 300 OULs. A color gradient from green to blue visually represents correlation strength, with green indicating lower correlation (0 to 0.5) and blue indicating higher correlation (0.5 to 1). The inset displays the intercorrelation histogram, where the similarity between any

two distinct OULs never exceeds 0.55. This high degree of uniqueness among the printed OULs is further supported by the inter-correlation distribution shown in the inset histogram, which approximately follows a normal distribution with the majority of probability mass concentrated between 0.32 and 0.55, yielding a mean of 0.39 and a standard deviation of 0.05.



(a) *FSIM* distribution of all 300 OULs. (b) Normalized *inter-HD* distribution.

Fig. 7. Evaluation of *FSIM* and *inter-HD*.

The binary responses used for the *inter-HD* evaluation were derived directly from the multi-scale CNN feature maps embedded in our authentication pipeline. Specifically, deep feature activations from ResNet-50 (layers 1 through 4) were flattened and concatenated into a robust 3,840-dimensional global descriptor. To ensure bit stability against noise, these continuous features were subsequently binarized using a median-centered guard band with a threshold of $\delta = 0.10$. Feature dimensions falling within this guard band margin were safely excluded as unstable components, yielding 2,867 highly reliable, effective bits per label for the final cryptographic evaluation. As illustrated by the empirical distribution in Fig 7(b), the computed normalized *inter-HD* yields an average value of 0.499 with a satisfactory empirical standard deviation of 0.17. This experimental result deviates from the ideal value of 0.5 by a negligible 0.10%. This exceptional performance strongly demonstrates that the microstructural features generated during the ink-sintering process exhibit excellent spatial randomness, rendering different labels highly distinct from one another.

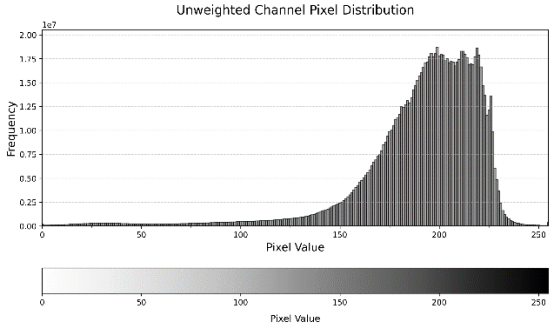


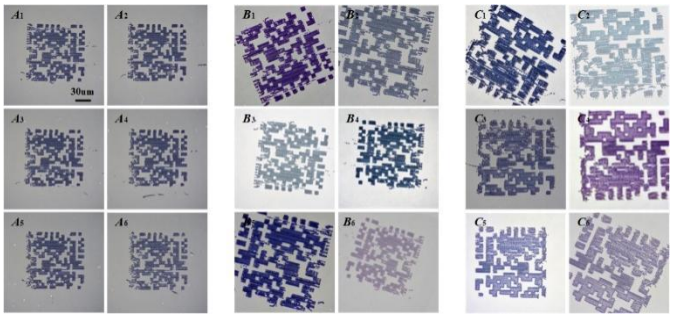
Fig. 8. Histogram of pixel values distributed in unweighted channels.

Each OUL pattern comprises 800×800 pixels, divided into 50×50 arrays, which are further subdivided into 4×4 subunit arrays each. Accordingly, the parameters L and R in (19) are calculated to be 4 and 200, respectively. Fig. 8 shows the histogram of pixel values distributed across the unweighted channels for all 300 OULs. It indicates that pixel values are predominantly distributed between 106 and 236, resulting in $C = 131$. A threshold of 187 is set to extract OUL physical features using ImageJ. The pixel filling ratio D, computed by applying the Otsu algorithm for OUL normalization, ranges from 0.4 to 0.5. Using $D = 4/9$ as an example, the encoding capacity of the OUL is calculated by the Carro-Temboury model to be 1.59×10^{3413}. This exceptionally high value substantially exceeds the latest nominal encoding capacity of 1.7×10^{1956} reported in perovskite fluorescent labels [38].

B. Results of AI Authentication System

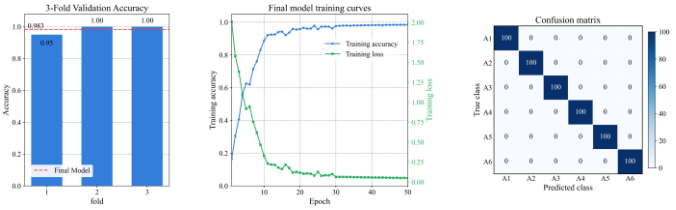
For fair comparison with prior works [27], [28], [29], we first selected six OULs from the original database, designated as A_1 to A_6, as shown in Fig. 9(a). Six augmented images, labeled B_1 to B_6, were generated from label A_1 by varying brightness, contrast, exposure, rotation angles, or a combination of multiple transformations, as illustrated in Fig. 9(b). In addition, six other OULs from the same 300 original OUL database were selected and denoted as C_1 to C_6, as shown in Fig. 9(c). These samples serve as same-process, non-enrolled OULs for open-set rejection evaluation.

Each OUL in Fig. 9(a) was captured 10 independent photos under varying conditions, and then was rotated at 3.6° intervals to produce 1,000 images per label, yielding a total of 6,000 samples. All images underwent uniform preprocessing, including outlier removal and format normalization. To reduce training time and improve generalization, model parameters were initialized using a pre-trained model and fine-tuned on our dataset. This transfer learning strategy accelerated convergence and enhanced model robustness [50].



(a) Original OULs. (b) Augmented OULs. (c) Faked OULs.

Fig. 9. Different types of OULs for authentication.



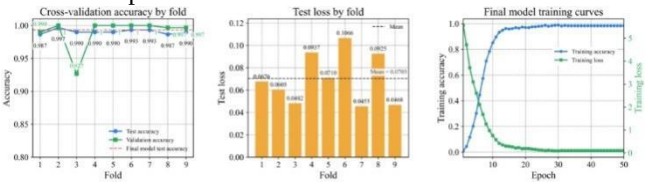
(a) Cross-validation accuracy (b) Final model training curves (c) Confusion matrix

Fig. 10. Model training and cross-validation performance of the six-class OULs

Three-fold cross-validation was used to evaluate the model. As shown in Fig. 10(a), the validation accuracies of each fold were 0.950, 1.000, and 1.000, respectively, while the final model achieved 0.983 (red dotted line). Final training was

completed over 50 epochs, lasting 30 minutes. Fig. 10(b) shows the final model training curves, where the blue and green solid lines represent training accuracy and loss, respectively. The accuracy increased rapidly during the first 20 epochs and subsequently stabilized near 1, whereas the loss gradually decreased toward zero. Fig. 10(c) shows the confusion matrix on the test set, all 600 test images, comprising 100 images per class, were correctly classified, resulting in a test accuracy of 100%. To evaluate the trained model, images B_1 to B_6 were used for authentication and were correctly classified into their corresponding classes, with the softmax probabilities of 0.969, 0.996, 0.977, 0.999, 0.993, and 0.997, respectively. Notably, the model exhibits strong generalization capability, as the six augmented images were not included in the training set. We also input the non-enrolled labels C_1 to C_6 into the model, and they were all correctly rejected, with the softmax probabilities below 0.5 in all cases, confirming the model's ability to reliably distinguish non-enrolled labels from genuine ones.

To further validate the AI model's applicability in complex scenarios, such as more categories and diverse images, we analyzed all 300 printed OULs and used them for model training. According the training process in Section IV, $G = 300$, $f = 10$, and $\alpha = 1000$. We captured 10 photos for each of 300 genuine labels and trained each model M_i ($i \in \{1,\ldots,9\}$) using the other 8 captures and their 1000 augmented derivatives as the training data with P_i as a validation set. The final model was obtained by training on P_1 to P_9 and their derivatives with the optimal training parameters among 9 trained models. Finally, P_{10} is used as the test set. With this increased data volume, the overall training requires approximately 50 hours to converge over 50×10 epochs.



(a) Accuracy of cross-validation. (b) Test loss of cross-validation. (c) Final model training curves.

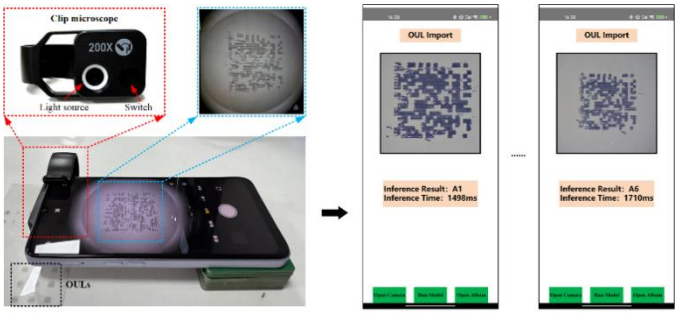
Fig. 11. Cross-validation and model training of all 300 OULs.

As shown in Fig. 11(a), the green and blue lines represent the validation and testing accuracy of each sub-model, respectively, while the red dashed line denotes the testing accuracy of the final model. The testing accuracies of the nine folds remain consistently high at around 0.990, with the final evaluation model reaching 0.993. As shown in Fig. 11(b), the testing losses across the nine folds remain consistently low, with the majority falling below 0.1. The minor variation in the 6th fold does not impact overall performance, yielding a highly satisfactory average testing loss of 0.070. Fig. 11(c) depicts the accuracy (blue curve) and loss (green curve) during the training process of the final model, stabilizing at approximately 0.984 and 0.092, respectively, after 30 epochs.

Following the Neyman–Pearson framework [51], we set the allowable error tolerance at $\alpha = 0.05$. The confidence threshold τ_p is determined as the 5th percentile of the maximum softmax probability distribution over correctly classified training samples, while the distance threshold τ_d is set as the 95th percentile of the within-class cosine distance distribution among known samples. This yields $\tau_p = 0.87$ and $\tau_d = 0.04$.

Utilizing these operational thresholds on our expanded dataset, the system achieves a 100% *TRR* across 200 unseen, identical-process genuine labels, while maintaining 97% classification *accuracy* on the enrolled test set. This experimental setup directly emulates a real-world deployment scenario where an authentication system encounters high-fidelity, unenrolled labels, providing strong empirical evidence of our scheme's practical robustness.

For practical product authentication, users can employ the Smart-ID mobile application to photograph OUL labels. As shown in Fig. 12(a), the system is fully integrated into smartphones, with a 200× clip-on microscope attached for enhanced imaging. The application returns a prediction value that determines whether the label is authentic or fake. Fig. 12(b) displays the authentication interface showing authentication results for A_1 to A_6. The AI model deployed for local inference operates as a black box, preventing attackers from accessing design details or tampering with the authentication logic.



(a) hardware (b) software

Fig. 12. The OUL authentication in the Smart-ID application.

Table IV compares the proposed scheme with recent PUF-based authentication approaches. Existing schemes exploit various physical randomness sources, including quantum dots [27], structural colors [28], [29], perovskites [38], and surface plasmons [40]. They generally require specialized functional materials, non-standard fabrication processes, or dedicated characterization instruments. For instance, He et al. [28] and Wei et al. [40] require spray coating and shadow deposition, respectively, while Han et al. [38] leverage perovskite crystallization. Schemes [38] and [40] also involve capital-intensive equipment, such as electron-beam evaporators or nanosecond UV laser engraving systems. In contrast, our OUL scheme uses commercially available AgNP ink and PET substrates and is fabricated through standard inkjet printing, offering better process compatibility and cost efficiency.

During authentication, schemes [28], [29] and [40] require optical microscopes, spectrum analyzers, or other laboratory-grade instruments, limiting their use in field or consumer-side scenarios. Our scheme instead implements lightweight AI-driven open-set authentication on a standard smartphone platform, enabling accessible and real-time verification. In terms of category and performance, most existing schemes are evaluated on a limited number of classes. Liu et al. [27], He et

al. [28], and Li et al. [29] consider only 6 label classes, while Wei et al. [40] consider only 2 classes. Although Han et al. [38] extend the number of classes to 60 and achieve an accuracy of 0.996, its fabrication cost remains high. By comparison, our scheme supports 300-class authentication with an *accuracy* of 0.970, a *TRR* of 100%, and an authentication time below 1s.

Benefiting from the stochastic spatial distribution of ink droplets, the proposed OUL achieves an encoding capacity of 1.59×10^{3413}, which is higher than those of the compared schemes. Overall, our proposed method achieves a favorable balance among low-cost fabrication, large-scale authentication, real-time verification, device accessibility, and encoding capacity. Table IV compares the proposed scheme with other recent authentication approaches.

The OUL fabrication process leverages standard inkjet printing on a flexible PET substrate, requiring no specialized functional compounds or custom manufacturing infrastructure. This enables seamless, low-overhead integration with existing PE workflows. The raw material consumption per OUL is estimated at approximately 0.11 cents. This is derived from a PET substrate of 0.1 cents per gram and commercial nano-silver ink priced at $22 per mL (at a concentration of 20 mg/mL). Given that each OUL comprises approximately 5,000 droplets with an individual volume of 10 pL, a single milliliter of ink yields roughly 2×10^4 labels, rendering per-unit substrate consumption negligible. On the computational side, initial model training requires approximately 50 hours of compute time and 97.66 MB of memory. Once trained, local edge inference and open-set verification can complete in under 1 second per authentication event, a processing latency ideally suited for real-time field deployments.

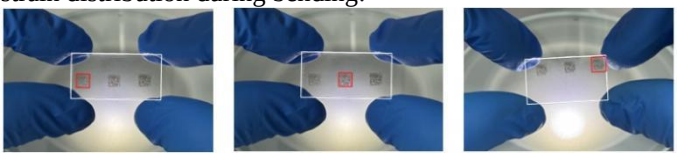
TABLE IV. COMPARATIVE MATRIX OF PUF-BASED AUTHENTICATION SCHEMES.

Schemes	Fabrication				Authentication						Encoding
	Methods	Material	Process	Cost	Network model	Classes	Data augment.	Time	Accuracy	Ease of use	
Liu et al. [27]	quantum dot	RGB QDs /PMMA	Inkjet printing	Low	AlexNet	6	single	≈2s	0.944	High	4.7×10^{202}
He et al. [28]	structure color	APS /SiO_2 MSs	Spray coating	Low	ResNet18	6	single	Few sec.	0.933	Medium	NA
Li et al. [29]	structure color	P(St-MMA-AA) NSs	Inkjet printing	Low	ResNet18	6	single	1~2s	>0.95	Low	NA
Han et al. [38]	perovskite	$CsPbBr_3$ /PDMS	Crystallization	High	PaCoNet	60	multiple	<1s	0.996	High	1.7×10^{1956}
Wei et al. [40]	surface plasmon	AuNPs /PDMS	Shadow deposition	High	ResNet18	2	multiple	NA	0.917	Medium	10^{816}
Proposed	ink droplets	AgNPs /PET	Inkjet printing	Low	ResNet50	300	multiple	<1s	0.970	High	1.59×10^{3413}

C. Reliability Analysis

1) Bending Reliability and Robust Encapsulation

To evaluate bending reliability under realistic conditions, we printed 4×4 mm^2 OULs on larger 3×1 cm^2 PET strips to simulate practical integrated PE devices. Within each strip, the OUL was positioned at three representative locations, center, edge, and corner, as shown in Fig. 13, to capture the effect of strain distribution during bending.



(a) Location at edge. (b) Location at center. (c) Location at corner.

Fig. 13. Bending reliability test configurations.

We evaluate bending performance under two scenarios. (1) Normal bending, where 50 samples were repeatedly bent to random angles between 0° to 120° for 200 cycles, with the samples allowed to spring back after each deformation to simulate typical handling or moderate mechanical disturbance; and (2) Extreme bending, where 25 samples were bent to induce a permanent bending set, resulting in a visible crease across the pattern to assess the upper bound of the OUL's tolerance. After the bending tests, all samples were recovered to a flat state for optical measurement.

The *FSIM* values of the OUL samples, computed by comparing post-bending images to their pre-bent baselines, are summarized in Table V.

TABLE V. *FSIM* OF OUL SAMPLES BEFORE/AFTER BENDING.

Location	Normal Case			Extreme Case		
	FSIM			*FSIM*		
	Num.	Mean	Std.	Num.	Mean	Std.
Center	16	0.95	0.06	5	0.85	0.05
Edge	17	0.94	0.08	10	0.89	0.06
Corner	17	0.95	0.06	10	0.87	0.09
Average	--	0.95	0.06	--	0.89	0.07

The high average *FSIM* value of 0.95 demonstrates that the printed OUL features remain exceptionally stable under normal cyclic bending. Even under the extreme crease condition, which represents a severe over-test, the OUL maintains an acceptable structural similarity (average *FSIM* of 0.89). Most importantly, all 75 tested samples were successfully authenticated, indicating that the patterns are robust against the mechanical variations expected in practical use.

To further safeguard the printed patterns against physical tampering, such as scratching, chemical treatment, or label removal, a transparent protective encapsulation layer can be integrated. Applying a thin coating of Parylene-C or an optically clear UV-curable polymer over the OUL enhances

mechanical robustness and prevents pattern delamination without compromising optical readability [52].

2) Environmental Reliability and Long-Term Aging Stability

To thoroughly evaluate the operational reliability of the printed OULs under real-world ambient stressors, we conducted systematic environmental exposure tests tracking performance variations under thermal stress (T), humidity fluctuations (H), and temporal degradation (A). *FSIM* was utilized to quantitatively assess the structural fidelity (Acc.) between the OUL images captured after stress exposure and their original pre-exposure baseline states. The results are shown in Table VI.

Thermal Stability: A cohort of 30 distinct OUL samples was evaluated across a temperature sweep from –20°C to 70°C at 10°C increments (with the baseline reference recorded at 30°C, 45% RH). Across the entire thermal operational range (spanning 300 discrete evaluation queries), the OUL features maintain a high average *FSIM* of 0.92 ± 0.12, with an authentication *accuracy* of 99.67%.

Humidity Sensitivity: Humidity testing was conducted by exposing 30 samples to relative humidity levels ranging from 25% to 85% RH in steps of 10% at room temperature. The micro-morphological patterns demonstrate excellent resilience against moisture, yielding a stable average *FSIM* of 0.96 ± 0.04 and an overall authentication *accuracy* of 99.52% across all testing points.

Short- and Long-Term Aging: For short-term temporal stability, 135 printed samples were monitored daily over a 14-day period under standard ambient laboratory conditions, yielding a consistent average *FSIM* of 0.94 ± 0.08 and a perfect 100% authentication *accuracy*. For long-term aging characterization, the 135 OUL samples stored a year ago were re-evaluated under ambient exposure. Although surface oxidation and dust accumulation reduced the average *FSIM* to 0.76 ± 0.04, the deep multi-scale features captured by our authentication network remained highly distinguishable, allowing 100% *accuracy* of the 1-year-aged OULs to successfully pass the enrollment verification criteria.

These findings collectively confirm that the printed silver-nanoparticle microstructural formations are highly stable against thermal and moisture variations while possessing strong intrinsic anti-aging characteristics that satisfy practical engineering requirements.

TABLE VI. ENVIRONMENTAL RELIABILITY

	Base	Num	step	Range	*FSIM*		Acc. (%)
					Mean	Std.	
T	30°C, 45% RH	30	10°C	–20~ 70°C	0.92	0.12	99.67
H	30°C, 45% RH	30	10%	25%~ 85% RH	0.96	0.04	99.52
A	30°C, 45% RH	135	1 D	1~14 D	0.94	0.08	100
		135	1 Y	1 Y	0.76	0.04	100

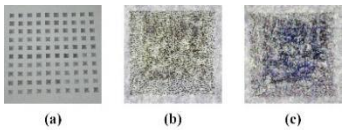
D. Security Analysis

In practical deployments, an adversary may launch various physical or digital attacks to compromise, fake, or bypass the proposed authentication framework. To evaluate system security, we establish a formal threat model and rigorously test our pipeline against three primary attack vectors.

1) Illegal Physical Cloning Attacks

We investigate three plausible strategies an attacker may adopt to physically clone an OUL:

(a) Naive Printing Replicas. An attacker may capture macroscopic or microscopic images of genuine labels and reprint them as grayscale or color replicas. To successfully deceive our verification model, the reproduction must preserve the microscopic printing artifacts that serve as high-entropy features. This requires a printer and scanner resolution exceeding 800 pixels per 4 mm, corresponding to more than 5,000 ink droplets per inch with individual droplet diameters under 5 μm. In contrast, standard commercial office printers typically produce droplets with diameters ranging from 50 μm to 100 μm, yielding an effective resolution of only 300 to 600 dots per inch (DPI). To substantiate this defense empirically, we physically printed a validation set of 100 cloned samples using several commonly available office printers (Fig. 14(a)). Microscopic imaging under both black-and-white and color modes (Fig. 14(b) and Fig. 14(c)) reveals a severe loss of microstructural integrity. The average *FSIM* between these naive clones and their genuine baselines drops to 0.2896, confirming that crude reprints exhibit fundamentally different microstructural patterns. Consequently, when submitted to our AI authentication pipeline, all 100 forged samples were successfully rejected with 100% *TRR*.



(a) Fabricated naive clones. (b) Black-and-white micro-image of a clone. (c) Colored micro-image of a clone.

Fig. 14. Micrographs showing structural degradation under a naïve physical cloning attack.

(b) Replicating the Same Printing Protocol. An attacker might attempt to replicate the labels by strictly duplicating the exact fabrication settings described in our work. To evaluate this scenario, we produced an independent batch of 200 physical labels under identical nominal printer settings. These labels were kept entirely separate from the original 300-class training distribution. When submitted to the verification system, all 200 batch-separated labels were successfully rejected with 100% *TRR*. This perfect rejection rate confirms that the underlying model effectively isolates intrinsic, process-specific micro-variations. Since these physical variations are governed by stochastic droplet coalescence during sintering, they are inherently non-repeatable across different fabrication batches.

(c) Advanced Precision Manufacturing. A highly sophisticated adversary might employ high-end micro/nano fabrication tools, such as femtosecond laser micromachining [53] or electrohydrodynamic (EHD) printing [54], which achieve feature resolutions at the micrometer or nanometer scale. Economically, this approach is highly unjustified for mass-market, low-cost applications, given that high-end

equipment costs range from several hundred thousand to over one million US dollars with hourly operating fees exceeding hundreds of dollars, whereas a genuine OUL costs approximately 0.11 cent per unit. Functionally, micro-scale fabrication remains inherently bound to stochastic process variations, including laser power fluctuations, ink droplet scattering, and substrate inhomogeneity. These inevitable microscopic variations introduce completely new, unclonable physical fingerprints rather than copying the target label, making deterministic structural replication at the feature level technically infeasible.

2) Machine Learning-Based Attack

An attacker may leverage machine learning techniques to mount sophisticated forgery attacks against OUL framework. We investigated two representative modern attack paradigms:

(a) Data-Driven Generative Forgery. In this scenario, the attacker is assumed to have collected a substantial dataset of genuine OUL samples. Specifically, they acquire a fraction of the 300 genuine OUL instances and capture multiple high-resolution images under controlled conditions. To augment the training set, they apply an extensive series of affine and photometric transformations-including rotation, translation, scaling, brightness adjustment, and mild Gaussian blurring. A Generative Adversarial Network (GAN) is then trained to approximate the underlying visual distribution of legitimate OUL fingerprints.

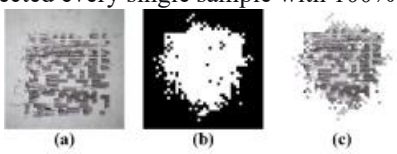
Let the number of collected genuine labels be 300r, where $r \in \{0.1,0.2,\ldots,0.9\}$ denotes the acquisition ratio. The attacker's objective is to synthesize a novel image capable of being accepted as any registered identity within the system. For each acquisition ratio r, the GAN was trained for 100 epochs and used to generate 1,000 synthetic digital samples. An attack is defined as successful if at least one generated image is accepted by the authentication system under the established open-set decision thresholds τ_p and τ_d.

Empirical results indicate that exactly zero successful forgeries were observed across the entire range of r from 0.1 to 0.9. This confirms that while the generative model can capture low-level macro-structural statistics, the synthesized samples remain far outside the narrow, process-specific feature boundaries required for positive enrollment verification.

(b) Black-Box Evasion (Adversarial Optimization). Under this threat model, the attacker is denied access to the internal network architecture, weights, or gradient information of the authentication system. However, they are permitted to submit candidate digital images and observe the binary verification outcomes indicating whether a sample matches a specified target label q. Starting from a bona fide target observation I_q, the attacker iteratively applies spatial masks and modifies localized image regions guided entirely by the returned verification feedback. Their goal is to isolate a minimum sufficient digital mask M^* that preserves just enough semantic information to trigger a positive verification decision.

As illustrated in Fig. 15(a), by executing over 2,000 continuous sequential queries, the optimization converged to a digital mask of minimal area (Fig. 15(b)) that is digitally accepted by the network as a match for the target pattern (Fig. 15(c)). It must be noted, however, that such high query complexity is often completely infeasible in practical deployments due to standard API rate-limiting and automated auditing mechanisms.

More critically, our experiments reveal that this digital-domain success does not generalize to the physical realm due to a phenomenon we term manufacturing sensitivity. The optimized template P^* (Fig. 15(c)) exhibits fundamental physical fragility upon re-printing. As demonstrated in our naive cloning experiments, even when employing a high-resolution printer (e.g., 1200 DPI), the stochastic micro-variations inherent in ink diffusion and substrate interaction introduce uncontrollable feature-space deviations. These physical deviations far exceed the strict verification tolerance ($\tau_d = 0.04$) required by the enrollment distribution. Over 100 independent physical reprinting trials of these optimized digital templates, the physical authentication model correctly rejected every single sample with 100% *TRR*.



(a) Target enrollment label. (b) Optimized digital mask. (c) Resulting physical printable pattern.

Fig. 15. Black-box adversarial optimization evasion attack and its physical reconstruction.

3) Authentication System and Software-Level Tampering

Apart from targeting the physical label or the machine learning model, an adversary might attempt to tamper directly with the execution environment of the AI authentication software. The objective of such an attack is typically to disrupt the verification pipeline or cause a denial-of-service (DoS) by preventing legitimate labels from being validated.

While code injection or reverse-engineering is technically feasible if the authentication software is deployed as a publicly accessible, unencrypted client application, this represents a general application security challenge rather than an intrinsic vulnerability of the OUL physical architecture. To systematically mitigate software-level tampering in practical applications, the authentication framework is designed to operate under a secure client-server architecture. By hosting the trained verification network on an encrypted, firewalled cloud server and exposing only a secure API endpoint to end-users, the internal parameters and decision boundaries of the AI model remain completely inaccessible to local tampering attempts. Furthermore, deploying code-obfuscation techniques and digital signature verification for mobile application clients ensures a robust, defense-in-depth security posture against software manipulation.

VI. Conclusion

PE products have become increasingly popular in recent years. However, their lightweight design makes them vulnerable to counterfeiting. To protect the copyright of PE products, we propose to implement an OUL by printing aqueous

AgNPs ink on PET film to form a unique pattern. This fabrication method is compatible with the standard PE printing process. Both the ink and PET substrate are cost-effective and readily accessible. The label can be produced in an area of 4×4 mm^2 while offering an encoding capacity of 1.59×10^{3413}, making it extremely difficult for counterfeiters to forge the label.

For authentication, we used ResNet-50 to train the model. The accuracy of unique authentication among six label categories outperforms existing methods while requiring minimum training time. To further validate the AI model's scalability and versatility, we enlarged to 300 printed OULs and used them for model training. The achieved accuracy is greater than 0.97. Besides reducing the error rate during the recognition process, the AI model is inherently more resistant to OUL counterfeiting attacks, as it is computationally expensive for attackers to reverse the model to understand its decision process than comparing the label against a database of stored reference values. Under bending, thermal, humidity, and extended aging conditions, the printed labels also maintain consistently accurate authentication.

This authentication system can be deployed on cloud or fog computing platforms, leveraging their superior computational resources to enable efficient product authentication for a large number of users via smartphone APIs.

While the proposed OUL-based AI authentication system demonstrates high performance and robust open-set security, several physical and algorithmic limitations must be acknowledged. First, resolving the 45 μm microstructural droplet features currently requires an inexpensive (< $20) 200× clip-on microscope accessory. While accessible, this introduces an external hardware dependency beyond a bare smartphone camera, and cross-device generalization across highly diverse mobile optical sensors remains to be formally validated in broader field studies. Second, regarding system scalability, our empirical validation was demonstrated on a 300-class active enrollment setup. When scaling the ecosystem to massive volumes (10^4-10^6 OULs), the system will require large-scale, high-dimensional vector search algorithms (e.g., approximate nearest neighbor search) within the feature-embedding space to maintain real-time matching efficiency without degrading retrieval accuracy. These considerations define the current operational boundaries and establish the trajectory for our ongoing research.

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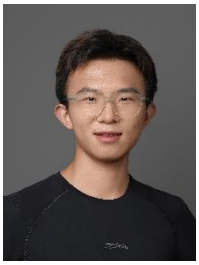
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